

Sentinel Code Review Report

File: sample1_simple.py | Language: Python | Analysis ID: validation_sample1_simple.py
Report generated: 2026-08-26 15:43:12 UTC

Executive / PR Summary

Analyzed sample1_simple.py (Python). No OWASP-standard security issues were flagged. The Code Analysis Agent flagged 1 structural issue(s).

PR Verdict: Approve with suggestions

Overall Code Health / Quality Score

Overall Score	Security	Quality	Maintainability	Complexity
96/100	100/100	92/100	96/100	Low

Severity Breakdown

Critical	High	Medium	Low	Info
0	0	1	0	0

Findings

■ [Medium] Mutable default argument

Category: Code Quality **Type:** Code Quality **Location:** Line 10

Explanation: `add\_item` (line 10) uses a mutable default argument — it is shared across every call.

Root cause: A list/dict default argument is created once at function-definition time and then shared across every call that doesn't pass its own value.

Recommended fix: Use `None` as the default and create the mutable object inside the function body.

Best practice: Default mutable arguments to None and construct the object inside the function body.

```
def f(x, items=None):  
    items = items if items is not None else []
```

Prioritized Remediation Roadmap

1. **[Medium] Mutable default argument** — Use `None` as the default and create the mutable object inside the function body.

Overall Code Quality / Security Assessment

Overall score 96/100. PR verdict: **Approve with suggestions**. 1 total finding(s) (0 critical, 0 high, 1 medium, 0 low).

Best practice suggestions:

- Default mutable function arguments to `None` and instantiate lists/dicts inside the function body.
- Leverage type annotations and context managers (`with` statements) for robust resource management.

Performance suggestions:

- Use generators, list comprehensions, and built-in algorithms for memory-efficient data processing.
- Utilize connection pooling and asynchronous I/O where applicable to maximize throughput.

Code smells: No significant code smells detected.